COMPUTER PROGRAMMING

Lab 13

LAB REPORT

TITLE

Structure in C++

OBJECTIVE

The students will learn and apply:

- Structures in C++
- Structure Members
- Structure Variables
- Accessing Structure Elements
- Initialization of Structures

SOFTWARE REQUIRED

- Dev-C++
- Code::Blocks
- C++ Compiler

THEORY

A structure is a collection of variables of different data types grouped together under one name.

The variables inside a structure are called structure members.

Structures can contain:

- int
- float
- char
- string

Unlike arrays, structures can store different data types together.

GENERAL SYNTAX

```
struct StructureName
{
```

```
    dataType member1;
    dataType member2;
};
```

LAB TASK 1

STRUCTURE DECLARATION AND INITIALIZATION

PROBLEM STATEMENT

Declare a structure called Person with:

- name
- age
- address

Initialize structure variable person1 and display values.

ALGORITHM

1. Start
2. Declare structure Person
3. Declare structure variable
4. Initialize members
5. Display values
6. End

PROGRAM

```
#include <iostream>
using namespace std;

// Structure declaration
struct Person
{
    string name;
    int age;
    string address;
};

int main()
{
    // Structure variable initialization
    Person person1;

    person1.name = "Aneesa Ibrahim";
    person1.age = 20;
    person1.address = "lower dir";
```

```
    // Displaying information
    cout << "Name: " << person1.name << endl;
    cout << "Age: " << person1.age << endl;
    cout << "Address: " << person1.address << endl;

    return 0;
}
```

OUTPUT

Name: Aneesa Ibrahim
Age: 20
Address: lower dir

EXPLANATION

- Person is a structure.
- person1 is structure variable.
- Dot operator . is used to access members.

FREE HAND DIAGRAM

```
        +----------------------+
person1 | Name    : Aneesa     |
        | Age     : 20         |
        | Address : lower dir  |
        +----------------------+
```

LAB TASK 2

STUDENT STRUCTURE

PROBLEM STATEMENT

Declare a structure called Student with:

- name
- age
- grade

Initialize student1 and display values.

ALGORITHM

1. Start
2. Declare structure
3. Create structure variable
4. Assign values
5. Display values

6. End

PROGRAM

```
#include <iostream>
using namespace std;

// Structure declaration
struct Student
{
    string name;
    int age;
    char grade;
};

int main()
{
    // Declaring structure variable
    Student student1;

    // Assigning values
    student1.name = "zaid Khan";
    student1.age = 20;
    student1.grade = 'A';

    // Displaying values
    cout << "Name: " << student1.name << endl;
    cout << "Age: " << student1.age << endl;
    cout << "Grade: " << student1.grade << endl;

    return 0;
}
```

OUTPUT

Name: zaid Khan
Age: 20
Grade: A

EXPLANATION

- Structure stores student information together.
- Members are accessed using dot notation.

FREE HAND DIAGRAM

```
         +------------------+
student1   | Name  : zaid Khan |
         | Age   : 20       |
```

| Grade : A |
+-------------------+

LAB TASK 3

ACCESSING STRUCTURE ELEMENTS

PROBLEM STATEMENT

Declare a structure called Book with:

- title
- author
- price
- pages

Create book1 and book2 and display their values.

ALGORITHM

1. Start
2. Declare structure Book
3. Create structure variables
4. Initialize values
5. Display values
6. End

PROGRAM

```
#include <iostream>
using namespace std;

// Structure declaration
struct Book
{
    string title;
    string author;
    float price;
    int pages;
};

int main()
{
    // Initializing structure variables
    Book book1, book2;

    book1.title = "C++ Programming";
    book1.author = "Bjarne";
    book1.price = 550.5;
    book1.pages = 350;
```

```
    book2.title = "Data Structures";
    book2.author = "Mark";
    book2.price = 700.0;
    book2.pages = 500;

    // Displaying book1
    cout << "Book 1 Information" << endl;
    cout << "Title: " << book1.title << endl;
    cout << "Author: " << book1.author << endl;
    cout << "Price: " << book1.price << endl;
    cout << "Pages: " << book1.pages << endl;

    // Displaying book2
    cout << "\nBook 2 Information" << endl;
    cout << "Title: " << book2.title << endl;
    cout << "Author: " << book2.author << endl;
    cout << "Price: " << book2.price << endl;
    cout << "Pages: " << book2.pages << endl;

    return 0;
}
```

OUTPUT

Book 1 Information
Title: C++ Programming
Author: Bjarne
Price: 550.5
Pages: 350

Book 2 Information
Title: Data Structures
Author: Mark
Price: 700
Pages: 500

EXPLANATION

- book1 and book2 are structure variables.
- Each variable stores book information separately.

RESULT

The programs were executed successfully and the concepts of structures and structure members were understood.

CONCLUSION

In this lab:

- We learned structures in C++.
- We created structure variables.
- We accessed structure members using dot operator.
- We stored multiple types of data in one structure.

Structures help organize related data efficiently.